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Homepage
Google Scholar
Semantic Scholar
1,700 Citations | h-index 12



Haoran Wang

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Research Interests

My research focuses on building trustworthy foundation models, with an emphasis on improving factuality, safety, and privacy through test-time control. I develop inference-time methods for large language models and multimodal systems that enhance robustness, interpretability, and reliability without retraining. I am broadly interested in interdisciplinary collaborations that translate trustworthy AI research into high-impact applications such as automated fact-checking, agentic systems, and AI for social good.

Education

Present **PhD, Computer Science**, *Emory University*, Atlanta, GA, USA.
Dissertation: Test-Time Approaches for Trustworthy Large Language Models
Advisor: Dr. Kai Shu, Committee Member: Dr. Li Xiong, Dr. Carl Yang, Dr. Xiangliang Zhang

2021 **MS, Computer Science**, *University of Oregon*, Eugene, Oregon, USA.
Advisor: Dr. Thien Huu Nguyen

2019 **BS, Computer Science**, *Purdue University*, West Lafayette, IN, USA.

Fellowships & Awards

IEEE BigData Rising Star, 2026

Google Cloud Research Credits, Google, 2026

Excellence in Research Assistance Award, Emory University, 2026

Research Access Program Award, OpenAI, 2025

Provost Doctoral Fellowship, Stevens Institute of Technology, 2022

Book Chapters and Journal Papers

[J1] ***Haoran Wang***, Xiongxiao Xu, Philip S. Yu, Kai Shu. Beyond Tokens: A Survey on Decoding Methods for Large Language Models and Large Vision-Language Models. *ACM SIGKDD Explorations Newsletter, Volume 28, Issue 1 (2026)*.

[B2] ***Haoran Wang***, Baixiang Huang, Kai Shu. Automated Fact-Checking. Chapter in *Oxford Handbook of Misinformation and Disinformation*, Oxford University Press.

[B1] Baixiang Huang, ***Haoran Wang***, Kai Shu. Factuality of Large Language Models: An Adversarial Perspective. Chapter in *Online Trust and Safety: Tools to Combat Online Harms, Misinformation, and Malicious Content*, Taylor and Francis CRC Press.

Conferences

[C14] Weizhi Zhang et al., including ***Haoran Wang***. JARVIS-Bench: Benchmarking Personal Intelligence Agents on Long-Horizon Real-User Daily Traces. In Proceedings of *The Fortieth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track (NeurIPS 2026)*.

[C13] ***Haoran Wang***, Li Xiong, Kai Shu. Do LLMs Know What Is Private Internally? Probing and

Steering Contextual Privacy Norms in Large Language Model Representations. *Proceedings of the Third Conference on Language Modeling (COLM 2026)*.

[C12] ***Haoran Wang***, Xiongxiao Xu, Baixiang Huang, Kai Shu. Privacy-Aware Decoding: Mitigating Privacy Leakage of Large Language Models in Retrieval-Augmented Generation. *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026)*.

[C11] Yue Huang, Chujie Gao, Siyuan Wu, ***Haoran Wang***, et al. TrustGen: A Platform of Dynamic Benchmarking on the Trustworthiness of Generative Foundation Models. In *Proceedings of the Fourteenth International Conference on Learning Representations (ICLR 2026)*.

[C10] Xiongxiao Xu, ***Haoran Wang***, Yueqing Liang, Philip S. Yu, Yue Zhao, Kai Shu. Can Multimodal LLMs Perform Time Series Anomaly Detection? In *Proceedings of the ACM Web Conference 2026 (WWW 2026)*.

[C9] Baixiang Huang, Zhen Tan, ***Haoran Wang***, Zijie Liu, Dawei Li, Ali Payani, Huan Liu, Tianlong Chen, Kai Shu. Model Editing as a Double-Edged Sword: Steering Agent Ethical Behavior Toward Beneficence or Harm. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2026)*.

[C8] Yueqing Liang, Liangwei Yang, Chen Wang, Congying Xia, Rui Meng, Xiongxiao Xu, ***Haoran Wang***, Ali Payani, Kai Shu. Benchmarking LLMs for Political Science: A United Nations Perspective. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2026)*.

[C7] ***Haoran Wang***, Kai Shu. Spatial-Aware Visual Program Guided Reasoning for Answering Complex Visual Questions. In *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2025*.

[C6] Aman Rangapur, ***Haoran Wang***, Ling Jian, Kai Shu. Fin-Fact: A Benchmark Dataset for Multimodal Financial Fact Checking and Explanation Generation. *Companion Proceedings of the ACM Web Conference 2025 (WWW 2025)*.

[C5] ***Haoran Wang***, Aman Rangapur, Xiongxiao Xu, Yueqing Liang, Haroon Gharwi, Carl Yang, Kai Shu. Piecing It All Together: Verifying Multi-Hop Multimodal Claims. In *Proceedings of the 31st International Conference on Computational Linguistics (COLING 2025)*.

[C4] ***Haoran Wang***, Kai Shu. Trojan Activation Attack: Red-Teaming Large Language Models using Steering Vectors for Safety-Alignment. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM 2024)*.

[C3] Yue Huang*, Lichao Sun*, ***Haoran Wang****, Siyuan Wu*, Qihui Zhang*, et al. TrustLLM: Trustworthiness in Large Language Models. In *Proceedings of the Forty-first International Conference on Machine Learning (ICML 2024)* (* indicates equal contribution).

[C2] ***Haoran Wang***, Kai Shu. Explainable Claim Verification via Knowledge-Grounded Reasoning with Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*.

[C1] ***Haoran Wang***, Yingtong Dou, Canyu Chen, Lichao Sun, Philip S. Yu, Kai Shu. Attacking Fake News Detectors via Manipulating News Social Engagement. In *Proceedings of the ACM Web Conference 2023 (WWW 2023)*.

Preprints

[P5] Yuyang Luo, ***Haoran Wang***, Kai Shu. Query-Only Backdoor Attacks on Self-Evolving Skills via Trajectory Poisoning. *Preprint 2026*.

[P4] Baixiang Huang, Limeng Cui, Jiapeng Liu, ***Haoran Wang***, Zhuiyue Tan, Yutong Chen, Chen Luo, Yi Liu, Kai Shu. Towards Effective Model Editing for LLM Personalization. *Preprint 2026*.

[P3] ***Haoran Wang***, Maryam Khalid, Qiong Wu, Jian Gao, Cheng Cao. Confidence-Guided Fact-Checking with Large Language Models through Probabilistic Certainty and Consistency. *Preprint 2026*.

[P2] Aman Rangapur, ***Haoran Wang***, Kai Shu. Investigating Online Financial Misinformation and Its Consequences: A Computational Perspective. *Preprint 2023*.

[P1] Canyu Chen, ***Haoran Wang***, Matthew Shapiro, Yunyu Xiao, Fei Wang, Kai Shu. Combating Health Misinformation in Social Media: Characterization, Detection, Intervention, and Open Issues. *Preprint 2022*.

Grant Proposal Writing Experience

NSF: SaTC: *Countering Disinformation Risks in the Era of Large Language Models*.
My Role: Discussing research objectives and writing multiple proposal tasks in interventions for influence risks posed by disinformation generated by large language models.

DHS: CAOE: *Countering Misinformation in the Era of Large Language Models*.
My Role: Discussing research objectives and writing multiple proposal tasks in explainable fact-checking using large language models.

ASFOR: YIP: *Influence Narratives Defender: Leveraging Social Media Research for Detection, Interpretation, and Assessment of Influence Narratives*.
My Role: Formulating the research topic and writing proposal task, *Narrative Representation and Extraction*, *Narrative Detection and Interpretation*, and *Impact Assessment and Mitigation*.

Invited Talk

Beyond Alignment: Inference-Time Steering for Secure and Trustworthy LLMs, *Amazon AGI*, Sep 2026.

Towards Trustworthy Large Language Models: An Inference-time Perspective, *Emory University*, April 2026.

Automated Fact-Checking for Reliable LLM Judges, *Amazon AGI*, May 2025.

Combating Misinformation in the Age of LLMs, *Illinois Institute of Technology*, October 2023.

Research Experience

Graduate Research Assistant, *Emory University*, Jan 2025 – Present.
PI: Kai Shu

Applied Scientist Intern, *Amazon AGI*, Sep 2026 – Dec 2026.
Manager: Qun Liu

Applied Scientist Intern, *Amazon AGI*, May 2025 – Aug 2025.
Manager: Cheng Cao

Graduate Research Assistant, *Illinois Institute of Technology*, Aug 2022 – May 2024.
PI: Kai Shu

Undergraduate Research Assistant, *Purdue University*, Aug 2018 – May 2019.
PI: Yung-Hsiang Lu

Teaching Experience

CS 570: Data Mining, *Spring 2026*, Emory University.
Guest Lecturer

CS 585: Natural Language Processing, *Fall 2023*, Illinois Institute of Technology.
Guest Lecturer

CS 550: Advanced Operating Systems, *Fall 2024*, Illinois Institute of Technology.
Graduate Teaching Assistant

CS 211: Computer Science I, *Spring 2020, Winter 2021, Spring 2021*, University of Oregon.
Graduate Teaching Assistant

CS 212: Computer Science II, *Fall 2020*, University of Oregon.
Graduate Teaching Assistant

Mentoring

Iris Qiao, Undergraduate student at Emory University → M.S., Harvard University

Aman Rangapur, IIT MS student → Solutions Architect, Allen Institute for AI (Ai2)

Academic Service

Memberships: ACM, ACL

Session Chair: CIKM 2024

Area Chair: NeurIPS Position Track {2025, 2026}

Program Committee: NeurIPS {2025, 2026}, ICML {2025}, AAAI {2024, 2025, 2026, 2027}, KDD {2024, 2025, 2026}, ACL {2025,2026}, PAKDD{2025}, ACML {2025, 2026}, ICLR {2026, 2027}, COLM {2026}, WWW{2027}, DocInsights@EMNLP{2026}, LASS@CIKM{2026}, IJCAI {2027}, IASEAI {2027}

Journal Reviewer: Big Data Research, Neural Networks (NEUNET)

External Reviewer: SIGIR {2023, 2024}, WSDM{2023}, ICDM{2023}, PAKDD{2024}, SDM{2025}

Student Volunteer: ACM FAccT 2023

Open-source Software

TrustGen (contributor): A modular and extensible toolkit for comprehensive trust evaluation of generative foundation models, (100+ Github ⭐ stars)

TrustLLM (contributor): Trustworthy LLM Benchmark and Toolkit, (500+ Github ⭐ stars)

Fin-Fact (contributor): Multimodal Financial Fact-Checking Dataset, (Benchmark dataset for shared task at Financial Misinformation Detection workshop at COLING 2025)